

1. Functional Requirements

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Requirement

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The system shall allow an OpenClaw agent to register a background task with a durable identifier, description, owner session, start time, check-in interval, and target chat/session.

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The system shall persist tracked task state in a durable local file so tracking survives agent/session restarts.

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The system shall periodically evaluate active tasks using a scheduler independent of agent memory.

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The system shall detect at minimum these states when observable: running, completed, failed, stale/overdue, and unknown.

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The system shall send a proactive update when a task completes.

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The system shall send a proactive update when a task fails or becomes blocked/overdue.

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The system shall avoid duplicate notifications by recording the last sent update timestamp and status.

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R The system shall support a configurable check-in cadence, with an initial
- target of 2–3 minutes when feasible.

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R The system shall record evidence of updates sent and task state
- transitions.

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R The system shall work with OpenClaw background/delegated tasks,
- including spawned sessions or long-running local processes, to the extent
1 that observable identifiers are available.

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2. Non-Functional Requirements

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Requirement

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R The system shall not depend on a human operator manually requesting
- status.

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R The system shall be transparent about what is observed vs inferred vs
- unverified.

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The system shall fail safely: if a task cannot be observed, it should emit an

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- honest degraded-status update rather than false success.

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The system shall be simple enough to operate on the current Ubuntu VM running OpenClaw.

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The system shall preserve a low enough operational overhead to avoid materially increasing token or compute waste.

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3. Constraints

- Must integrate with OpenClaw as it exists today.
- Must not rely on fake reminders or unenforced agent promises.
- Must produce real, inspectable evidence.